

Number Sense

Integers · BEDMAS · Exponents · Fractions · Roots · Percent · Ratios

MTH1W · Class — Guided Practice

Name: _____

Date: _____

Goal: Use integers, exponent laws, fractions, roots, percents and ratios to solve multi-step problems.

Warm-up (do these first — recall)

Answer quickly to activate prior skills. Answers are on the last page.

1. Evaluate $(-4) + (-9)$.

2. Write $2 \times 2 \times 2 \times 2$ as a single power, then find its value.

3. Reduce the fraction $12/18$ to lowest terms.

Worked Examples & Guided Steps

For each skill: study the worked **Example** ('I do'), then complete the **We do** steps together, filling every blank.

Integers

Example:

Evaluate $(-7) - (-2) + (-4)$.

Rewrite subtraction as adding the opposite: $(-7) + (+2) + (-4)$.

Combine: $(-7 + 2) = -5$, then $(-5) + (-4) = -9$.

Answer: -9.

We do — Evaluate $(-9) + (+5) - (-6)$.

Rewrite: $(-9) + (+5) + (\underline{\quad})$.

Combine step by step: $-9 + 5 = \underline{\quad}$, then $\underline{\quad} + 6 = \underline{\quad}$.

Answer: .

Order of Operations (BEDMAS)

Example:

Evaluate $20 - 2 \times (3 + 1)^2$.

Brackets first: $(3 + 1) = 4$. Exponent: $4^2 = 16$.

Multiply: $2 \times 16 = 32$. Subtract: $20 - 32 = -12$.

Answer: -12.

We do — Evaluate $6 + 12 \div (5 - 2)^2$.

Brackets: $(5 - 2) = \underline{\quad}$. Exponent: $\underline{\quad}^2 = \underline{\quad}$.

Divide: $12 \div \underline{\quad} = \underline{\quad}$. Add: $6 + \underline{\quad} = \underline{\quad}$.

Answer: $\underline{\quad}$.

Exponent Rules

Example:

Simplify $(x^4 \times x^3) \div x^2$, $x \neq 0$.

Product rule (add): $x^{4+3} = x^7$.

Quotient rule (subtract): $x^{7-2} = x^5$.

Answer: x^5 .

We do — Simplify $(a^2)^3 \times a^0$, $a \neq 0$.

Power of a power (multiply): $a^{2 \times 3} = a^{\underline{\quad}}$.

Zero exponent: $a^0 = \underline{\quad}$.

Combine: $a^{\underline{\quad}} \times \underline{\quad} = a^{\underline{\quad}}$.

Fractions — Four Operations

Example:

Evaluate $3/4 \div 2/3 + 1/2$.

Divide: multiply by the reciprocal $\rightarrow 3/4 \times 3/2 = 9/8$.

Common denominator with $1/2 = 4/8$: $9/8 + 4/8 = 13/8$.

Answer: $13/8$ (or $1 \frac{5}{8}$).

We do — Evaluate $5/6 - 1/4 \times 2/3$.

Multiplication before subtraction: $1/4 \times 2/3 = \underline{\quad}$.

Common denominator for $5/6$ and that product: LCD = $\underline{\quad}$.

Subtract: $\underline{\quad} - \underline{\quad} = \underline{\quad}$.

Rational Numbers

Example:

Order these from least to greatest: -0.6 , $-2/3$, 0.5 , $-1/2$.

Write all as decimals: $-2/3 \approx -0.667$, $-1/2 = -0.5$.

Compare: $-0.667 < -0.6 < -0.5 < 0.5$.

Answer: $-2/3$, -0.6 , $-1/2$, 0.5 .

We do — Order from least to greatest: $-3/4$, -0.7 , $1/4$, -1 .

As decimals: $-3/4 = \underline{\hspace{1cm}}$, $1/4 = \underline{\hspace{1cm}}$.

Place on a number line and compare the negatives first.

Answer: $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$, $\underline{\hspace{1cm}}$.

Square Roots & Irrationals

Example:

Is $\sqrt{50}$ rational or irrational? Estimate it to one decimal place.

50 is not a perfect square, so $\sqrt{50}$ is irrational.

$49 = 7^2$ and $64 = 8^2$, so $\sqrt{50}$ is just above 7.

Estimate: $\sqrt{50} \approx 7.1$.

We do — Is $\sqrt{0.16}$ rational? Find its exact value.

$0.16 = 16/100$, and $\sqrt{(16/100)} = \underline{\hspace{1cm}} / \underline{\hspace{1cm}}$.

Simplify: $\underline{\hspace{1cm}} = \underline{\hspace{1cm}}$ (a terminating decimal).

So $\sqrt{0.16}$ is $\underline{\hspace{1cm}}$ (rational / irrational).

Fractions ↔ Decimals ↔ Percent

Example:

Convert $7/8$ to a decimal and a percent.

Divide: $7 \div 8 = 0.875$.

Percent: $0.875 \times 100 = 87.5\%$.

Answer: $0.875 = 87.5\%$.

We do — Convert $3/20$ to a decimal and a percent.

Make the denominator 100: $3/20 = \underline{\hspace{1cm}}/100$.

Decimal: $\underline{\hspace{1cm}}$. Percent: $\underline{\hspace{1cm}}\%$.

Answer: $3/20 = \underline{\hspace{1cm}} = \underline{\hspace{1cm}}\%$.

Applying Percents

4. Write 0.35 as a fraction in lowest terms, then as a percent. **[L1]**

5. Evaluate: $\frac{2}{3} - \frac{1}{4} + \frac{5}{6}$. Give the answer in lowest terms. **[L2]**
6. Simplify: $(2^3)^2 \times 2^0$. Give a single power of 2 and its value. **[L2]**
7. A jacket costs \$80. It is discounted 15%, then 13% HST is added to the sale price. Find the final price. **[L2]**
8. Between which two consecutive whole numbers does $\sqrt{55}$ lie? Justify with two perfect squares. **[L2]**
9. A recipe uses flour to sugar in the ratio 5 : 2. If 15 cups of flour are used, how much sugar is needed? **[L2]**

10. A car travels 240 km in 3 hours, then 150 km in 2.5 hours. Find the average speed for the whole trip. Explain why it is not the average of the two speeds. **[L3]**
11. Sam simplified $3^2 + 3^2$ as 3^4 . Explain the error and give the correct value. **[L3]**
12. A number increased by 25% gives 90. Work backwards to find the original number, and check your answer. **[L3]**

Guided Practice — Answers

Warm-up: 1) -13 2) $2^4 = 16$ 3) $2/3$

- 1.** (L1) -10
- 2.** (L1) $5^{6-2} = 5^4$
- 3.** (L1) $17(3 + 16 - 2)$
- 4.** (L1) $7/20$; 35%
- 5.** (L2) $5/4$
- 6.** (L2) $2^6 \times 1 = 2^6 = 64$
- 7.** (L2) Sale $80 \times 0.85 = \$68.00$; +13% $\rightarrow 68 \times 1.13 = \76.84
- 8.** (L2) 7 and 8, since $7^2=49 < 55 < 64=8^2$
- 9.** (L2) $15 \div 5 \times 2 = 6$ cups
- 10.** (L3) $(240+150)/(3+2.5) = 390/5.5 = 70.9$ km/h. It weights by distance/time, not a simple mean of 80 and 60.
- 11.** (L3) Error: exponents are only added when MULTIPLYING like bases. Here it is addition: $9 + 9 = 18$, not 81.
- 12.** (L3) $x \times 1.25 = 90 \rightarrow x = 90/1.25 = 72$. Check: $72 \times 1.25 = 90$ ✓